



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- Show all necessary working in Questions 11–14

Total Marks - 70 Marks

Section I 10 Marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II 60 Marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section.

Examiner: *External Examiner*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section I

10 marks

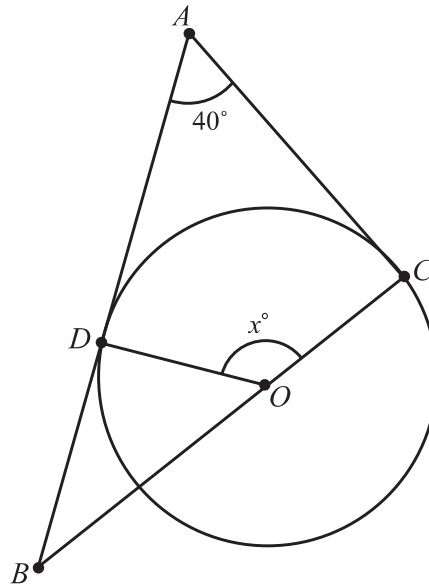
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 What is the expression $\frac{2 \tan A}{1 + \tan^2 A}$ equal to?
- (A) $\cos 2A$
(B) $\sin 2A$
(C) $\tan 2A$
(D) $\cot 2A$
- 2 The polynomial, $p(x)$ is defined by $p(x) = x^3 - x^2 + x + 3$.
What is the remainder when $p(x)$ is divided by $(x - 1)$?
- (A) 0
(B) 2
(C) 3
(D) 4
- 3 What is the domain and range of $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$?
- (A) $|x| \leq 3, |y| \leq \pi$.
(B) $|x| \leq 1, |y| \leq 3$.
(C) $|x| \leq 1, |y| \leq \pi$.
(D) $|x| \leq 3, |y| \leq 2$.
- 4 What ratio does the point $P(10, 11)$ divide the interval AB , where $A(-2, 3)$ and $B(7, 9)$?
- (A) 1 : 4
(B) 4 : -1
(C) 1 : -4
(D) 4 : 1

- 5 In the figure below, a circle with centre O is tangent to AB at point D and tangent to AC at point C .



If $\angle A = 40^\circ$, what is the value of x ?

- (A) 140
(B) 145
(C) 150
(D) 155
- 6 The function $f(x) = \sin x - \frac{2}{3}x$ has a real root close to $x = 1.5$.
Let $x = 1.5$ be a first approximation to the root.
What is the second approximation to the root using Newton's method?
- (A) 1.495
(B) 1.496
(C) 1.503
(D) 1.504
- 7 A test is administered with 15 questions. Students are allowed to answer any ten.
How many choices of ten questions are there?
- (A) 150
(B) 250
(C) 3003
(D) 3000

- 8** The graph of $f(x) = 0.6 \cos^{-1}(x-1)$, defines a curve that, when rotated about the y -axis, will produce a solid that is to be the shape and size of a new biscuit. Which integral expression will give the volume of the biscuit?

(A) $\pi \int_0^{0.6} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$

(B) $\pi \int_0^{0.6} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$

(C) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$

(D) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$

- 9** What is the value of $\lim_{n \rightarrow \infty} \left(n \sin \frac{\pi}{n} \right)$?

(A) $-\infty$

(B) 0

(C) π

(D) ∞

- 10** What is the x -intercept of the normal to the parabola $x^2 = 4ay$ at the point $(2ap, ap^2)$ on the parabola?

(A) $ap(p^2 + 1)$

(B) $ap(p^2 + 2)$

(C) ap^2

(D) $-ap^2$

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Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a NEW writing booklet. Extra pages are available

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Start a NEW Writing Booklet

- (a) Evaluate $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$ 2
- (b) Find the acute angle between the lines $y = \sqrt{3}x - 2$ and $y = -\sqrt{3}x + 1$. 2
- (c) The point $(-6t, 9t^2)$, where t is a variable, lies on a curve. 2
Find the Cartesian equation of the curve.
- (d) Use the substitution $u = x^4 + 2$ to evaluate $\int_0^1 \frac{x^7}{(x^4 + 2)^2} dx$, 3
leaving your answer in the form $p \ln q + r$.
- (e) Find $\frac{d}{dx}(x^2 \tan^{-1} x)$. 2
- (f) (i) Find a general solution of the equation 3
$$\cos\left(3x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

(ii) Hence, find the smallest solution of this equation which is greater than 5π . 1

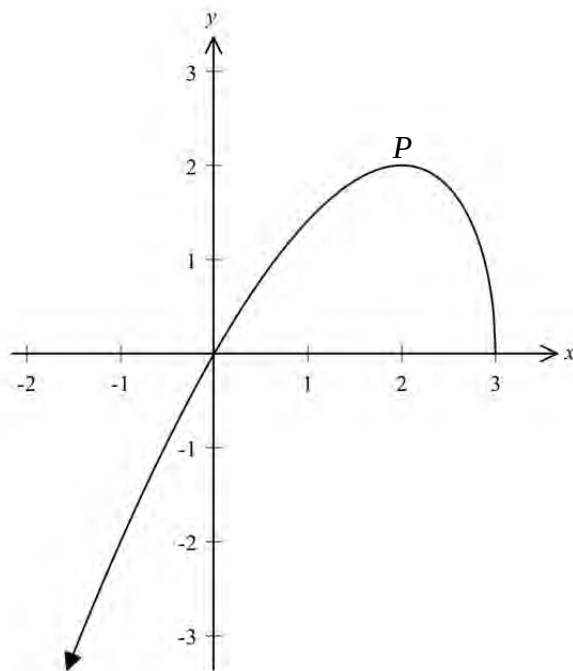
Question 12 (15 Marks) Start a NEW Writing Booklet

- (a) The cubic equation $x^3 + px^2 + qx + r = 0$, where p , q and r are real, has roots α , β and γ .
- (i) Given that $\alpha + \beta + \gamma = 4$ and $\alpha^2 + \beta^2 + \gamma^2 = 20$, find the values of p and q . 2
- (ii) Given further that one root is 4, find the value of r . 1
- (b) (i) Show that $(2\sin x + \cos x)^2$ can be written in the form 2
- $$\frac{5}{2} + 2\sin 2x - \frac{3}{2}\cos 2x$$
- (ii) Find the exact volume when the graph of $y = 2\sin x + \cos x$ for $0 \leq x \leq \frac{\pi}{2}$ is rotated about the x -axis. 2
- (c) A particle moves in simple harmonic motion about a fixed origin O with a period of $\frac{2\pi}{5}$ seconds. Initially, $x = 1$ and $\dot{x} = -5\sqrt{3}$.
- (i) Find x as a function of t . 3
- (ii) Find the first time that the particle passes back through $x = 1$. 2
- (d) Solve the inequality $\frac{x}{(x+1)(x-2)} \leq -\frac{1}{2}$ 3

End of Question 12

Question 13 (15 Marks) Start a NEW Writing Booklet

- (a) The diagram below shows the graph of the curve $y = x\sqrt{3-x}$.
 The point P has coordinates $(2, 2)$ and is a stationary point.
 The function $f(x)$ is defined by $f(x) = x\sqrt{3-x}$, $x \leq 3$.



- (i) On the same diagram, sketch the region satisfied by $y \leq f(x)$ and $y \geq f^{-1}(x)$ 3

- (ii) Explain why the area A of the shaded region in (i) is given by 2

$$A = \int_0^2 (x\sqrt{3-x} - x) dx$$

Do NOT attempt to evaluate this integral.

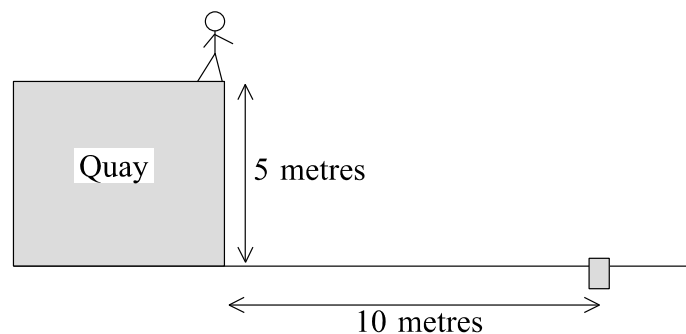
- (b) Nine different pies are to be divided between three people so that each person gets an odd number of pies.
 Find the number of ways this can be done. 2

- (c) Show that $\lim_{x \rightarrow 5} \frac{x^3 - x^2 - 100}{x - 5} = 65$ 2

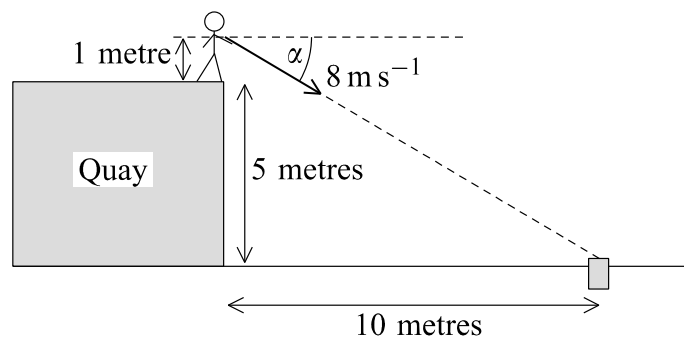
Question 13 continues on page 9

Question 13 (continued)

- (d) A girl stands at the edge of a quay and sees a tin can floating in the water. The water level is 5 metres below the top of the quay and the can is at a horizontal distance of 10 metres from the quay, as shown in the diagram.



The girl decides to throw a stone at the can. She throws the stone from a height of 1 metre above the top of the quay. The initial velocity of the stone is 8 m s^{-1} at an angle α below the horizontal, so that the initial velocity of the stone is directed at the can, as shown in the diagram below.



Assume that the stone is a particle and that it experiences no air resistance as it moves. The equations of motion of the stone are

$$x = 8t \cos \alpha \text{ and } y = 8t \sin \alpha - 4.9t^2. \text{ (Do NOT prove this.)}$$

- | | | |
|-------|---|----------|
| (i) | Find α .
Leave your answer correct to the nearest degree. | 2 |
| (ii) | Find the time that it takes for the stone to reach the level of the water. | 2 |
| (iii) | Find the distance between the stone and the can, when the stone hits the water. | 2 |

End of Question 13

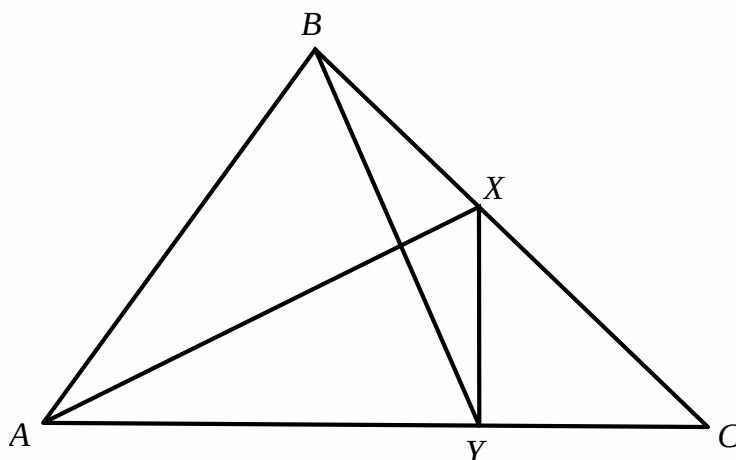
Question 14 (15 Marks) Start a NEW Writing Booklet

(a) (i) Show that $\frac{1}{(k+2)!} - \frac{k+1}{(k+3)!} = \frac{2}{(k+3)!}$, where k is an integer. 1

(ii) Prove by induction that, for all positive integers n , 3

$$\sum_{r=1}^n \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{n+1}}{(n+2)!}$$

(b) X and Y are points on the sides BC and AC of a triangle ABC respectively such that $\angle AXC = \angle BYC$ and $BX = XY$.



Copy or trace the diagram into your answer booklet.

(i) Show that $\angle XAC = \angle YBC$. 2

(ii) Hence, explain why $ABXY$ is a cyclic quadrilateral. 1

(iii) Prove that AX bisects the angle $\angle BAC$. 2

Question 14 continues on page 11

Question 14 (continued)

(c) If $\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$, and $x = 70$ when $t = 0$, find x as a function of t . **3**

(d) Liquid fuel is stored in a tank. At time t minutes, the depth of fuel in the tank is x cm. Initially there is a depth of 70 cm of fuel in the tank. There is a tap 6 cm above the bottom of the tank. The flow of fuel out of the tank is modeled by the differential equation

$$\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$$

(i) Explain what happens when $x = 6$. **1**

(ii) Find how long it will take for the depth of fuel to fall from 70 cm to 22 cm. **2**

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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Mathematics Extension 1

Sample Solutions

Section I

1 What is the expression $\frac{2 \tan A}{1 + \tan^2 A}$ equal to?

- (A) $\cos 2A$
 (B) $\sin 2A$
 (C) $\tan 2A$
 (D) $\cot 2A$

If $t = \tan \frac{\theta}{2}$ then $\sin \theta = \frac{2t}{1+t^2}$.

2 The polynomial, $p(x)$ is defined by $p(x) = x^3 - x^2 + x + 3$.

What is the remainder when $p(x)$ is divided by $(x-1)$?

- (A) 0
 (B) 2
 (C) 3
 (D) 4

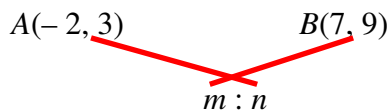
Remainder = $p(1) = 1 - 1 + 1 + 3 = 4$

3 What is the domain and range of $y = 2 \sin^{-1} \left(\frac{x}{3} \right)$?

- (A) $|x| \leq 3, |y| \leq \pi.$ $-\frac{\pi}{2} \leq \frac{y}{2} = \sin^{-1} \left(\frac{x}{3} \right) \leq \frac{\pi}{2} \Rightarrow -\pi \leq y \leq \pi$
 (B) $|x| \leq 1, |y| \leq 3.$
 (C) $|x| \leq 1, |y| \leq \pi.$ $-1 \leq \frac{x}{3} \leq 1 \Rightarrow -3 \leq x \leq 3$
 (D) $|x| \leq 3, |y| \leq 2.$

4 What ratio does the point $P(10, 11)$ divide the interval AB , where $A(-2, 3)$ and $B(7, 9)$?

- (A) 1 : 4
 (B) 4 : -1
 (C) 1 : -4
 (D) 4 : 1



$$10 = \frac{-2n + 7m}{m + n} \Rightarrow 10m + 10n = -2n + 7m$$

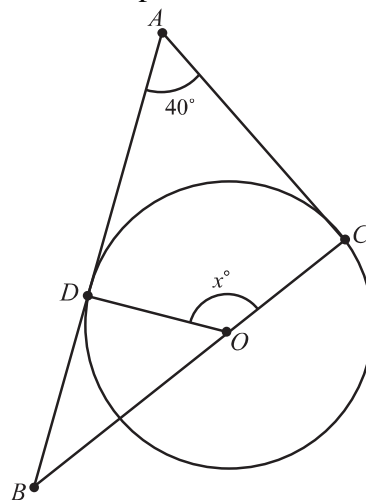
$$\therefore 12n = -3m \Rightarrow \frac{m}{n} = -4$$

- 5 In the figure below, a circle with centre O is tangent to AB at point D and tangent to AC at point C .

If $\angle A = 40^\circ$, what is the value of x ?

- (A) 140
(B) 145
(C) 150
(D) 155

$\angle ADO = \angle ACO = 90^\circ$ (radius and tangent)
 $\therefore x + 2 \times 90 + 40 = 360$ (angle sum $ADOC$)
 $\therefore x = 140$



- 6 The function $f(x) = \sin x - \frac{2}{3}x$ has a real root close to $x = 1.5$

Let $x = 1.5$ be a first approximation to the root.

What is the second approximation to the root using Newton's method?

- (A) 1.495
 (B) 1.496
 (C) 1.503
 (D) 1.504
- $$f'(x) = \cos x - \frac{2}{3} \quad x_1 = 1.5 - \frac{f(1.5)}{f'(1.5)}$$
- $$= 1.5 - \frac{\sin 1.5 - \frac{2}{3} \times \frac{3}{2}}{\cos 1.5 - \frac{2}{3}}$$
- $$= 1.496$$

- 7 A test is administered with 15 questions. Students are allowed to answer any ten. How many choices of ten questions are there?

- (A) 150
(B) 250
(C) 3003
(D) 3000
- $${}^{15}C_{10} = 3003$$

- 8 The graph of $f(x) = 0.6 \cos^{-1}(x-1)$, defines a curve that, when rotated about the y-axis, will produce a solid that is to be the shape and size of a new biscuit. Which integral expression will give the volume of the biscuit?

(A) $\pi \int_0^{0.6} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$ $y = \frac{3}{5} \cos^{-1}(x-1)$

(B) $\pi \int_0^{0.6} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$ $\therefore x = \cos\left(\frac{5}{3}y\right) + 1$

(C) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{3}{5}y\right) + 1 \right]^2 dy$ $0 \leq \frac{5}{3}y = \cos^{-1}(x-1) \leq \pi$

☒ (D) $\pi \int_0^{0.6\pi} \left[\cos\left(\frac{5}{3}y\right) + 1 \right]^2 dy$ $\therefore 0 \leq y \leq \frac{3}{5}\pi$

- 9 What is the value of $\lim_{n \rightarrow \infty} \left(n \sin \frac{\pi}{n} \right)$?

(A) $-\infty$

(B) 0

☒ (C) π

(D) ∞

$$\lim_{n \rightarrow \infty} \left(n \sin \frac{\pi}{n} \right) = \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{\pi}{n}}{\frac{1}{n}} \right)$$

$$= \lim_{u \rightarrow 0} \left(\frac{\sin \pi u}{u} \right) \quad \left[u = \frac{1}{n} \right]$$

$$= \pi$$

- 10 What is the x-intercept of the normal to the parabola $x^2 = 4ay$ at the point $(2ap, ap^2)$ on the parabola?

(A) $ap(p^2 + 1)$ $x + py = 2ap + ap^3$

☒ (B) $ap(p^2 + 2)$ $\therefore y = 0, x = 2ap + ap^3$

(C) ap^2

(D) $-ap^2$

Section II

Question 11

- (a) Evaluate $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$ 2

$$\begin{aligned}\int_0^1 \frac{dx}{\sqrt{2-x^2}} &= \left[\sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^1 \\ &= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ &= \frac{\pi}{4}\end{aligned}$$

- (b) Find the acute angle between the lines $y = \sqrt{3}x - 2$ and $y = -\sqrt{3}x + 1$. 2

$$m_1 = \sqrt{3}, m_2 = -\sqrt{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2\sqrt{3}}{1-3} \right|$$

$$= \sqrt{3}$$

$$\theta = 60^\circ$$

Accept an answer of $\theta = \frac{\pi}{3}$.

- (c) The point $(-6t, 9t^2)$, where t is a variable, lies on a curve. 2

Find the Cartesian equation of the curve.

$$x = -6t \Rightarrow t = -\frac{1}{6}x$$

$$\therefore y = 9t^2 = 9\left(-\frac{1}{6}x\right)^2$$

$$\therefore x^2 = 4y$$

- (d) Use the substitution $u = x^4 + 2$ to evaluate $\int_0^1 \frac{x^7}{(x^4 + 2)^2} dx$, 3

leaving your answer in the form $p \ln q + r$.

$$u = x^4 + 2 \Rightarrow du = 4x^3 dx$$

$$x = 0, u = 2$$

$$x = 1, u = 3$$

$$\begin{aligned}\int_0^1 \frac{x^7}{(x^4 + 2)^2} dx &= \frac{1}{4} \int_0^1 \frac{x^4 (4x^3 dx)}{(x^4 + 2)^2} = \frac{1}{4} \int_2^3 \frac{u-2}{u^2} du = \frac{1}{4} \int_2^3 \left(\frac{1}{u} - \frac{2}{u^2} \right) du \\ &= \frac{1}{4} \left[\ln u + \frac{2}{u} \right]_2^3 = \frac{1}{4} \left[\left(\ln 3 + \frac{2}{3} \right) - \left(\ln 2 + \frac{2}{2} \right) \right] = \frac{1}{4} \left(\ln \frac{3}{2} - \frac{1}{3} \right) \\ &= \frac{1}{4} \ln \frac{3}{2} - \frac{1}{12}\end{aligned}$$

Question 11 (continued)

- (e) Find $\frac{d}{dx}(x^2 \tan^{-1} x)$. **2**

$$\begin{aligned}\frac{d}{dx}(x^2 \tan^{-1} x) &= x^2 \times \frac{1}{1+x^2} + 2x \tan^{-1} x \\ &= \frac{x^2}{1+x^2} + 2x \tan^{-1} x\end{aligned}$$

- (f) (i) Find a general solution of the equation **3**

$$\cos\left(3x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$3x - \frac{\pi}{6} = 2n\pi \pm \cos^{-1}\left(\frac{\sqrt{3}}{2}\right), n \in \mathbb{Z}$$

$$\therefore 3x = 2n\pi \pm \frac{\pi}{6} + \frac{\pi}{6}$$

$$= 2n\pi + \frac{\pi}{3}, 2n\pi$$

$$\therefore x = \frac{2}{3}n\pi + \frac{\pi}{9}, \frac{2}{3}n\pi$$

- (ii) Hence, find the smallest solution of this equation which is greater than 5π . **1**

$$\frac{2}{3}n\pi + \frac{\pi}{9} \geq 5\pi \Rightarrow n \geq \frac{22}{3}$$

$$\therefore n = 8$$

$$\text{Smallest angle} = \frac{2}{3} \times 8\pi = \frac{16}{3}\pi$$

Question 12

- (a) The cubic equation $x^3 + px^2 + qx + r = 0$, where p , q and r are real, has roots α , β and γ .

- (i) Given that $\alpha + \beta + \gamma = 4$ and $\alpha^2 + \beta^2 + \gamma^2 = 20$, find the values of p and q . 2

$$p = -(\alpha + \beta + \gamma)$$

$$= -4$$

$$q = \alpha\beta + \beta\gamma + \gamma\alpha$$

$$\alpha^2 + \beta^2 + \gamma^2 = 20$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= 16 - 2q$$

$$\therefore q = -2$$

$$\therefore p = -4, q = -2$$

- (ii) Given further that one root is 4, find the value of r . 1

$$\therefore x^3 - 4x^2 - 2x + r = 0$$

$$\text{Substitute } x = 4: \quad 4^3 - 4 \times 16 - 2 \times 4 + r = 0$$

$$\therefore r = 8$$

- (b) (i) Show that $(2\sin x + \cos x)^2$ can be written in the form 2

$$\frac{5}{2} + 2\sin 2x - \frac{3}{2}\cos 2x$$

$$(2\sin x + \cos x)^2 = 2 \times 2\sin^2 x + 2 \times 2\sin x \cos x + \cos^2 x$$

$$= 2(1 - \cos 2x) + 2\sin 2x + \frac{1}{2}(1 + \cos 2x)$$

$$= \frac{5}{2} + 2\sin 2x - \frac{3}{2}\cos 2x$$

- (ii) Find the exact volume when the graph of $y = 2\sin x + \cos x$ 2

for $0 \leq x \leq \frac{\pi}{2}$ is rotated about the x -axis.

$$\text{Volume} = \pi \int_0^{\frac{\pi}{2}} (2\sin x + \cos x)^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(\frac{5}{2} + 2\sin 2x - \frac{3}{2}\cos 2x \right) dx$$

$$= \pi \left[\frac{5}{2}x - \cos 2x - \frac{3}{4}\sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\left(\frac{5\pi}{4} - \cos \pi - \frac{3}{4}\sin \pi \right) - (-1) \right]$$

$$= \frac{5\pi^2}{4} + 2\pi$$

Question 12 (continued)

- (c) A particle moves in simple harmonic motion about a fixed origin O with a period of $\frac{2\pi}{5}$ seconds. Initially, $x = 1$ and $\dot{x} = -5\sqrt{3}$.

- (i) Find x as a function of t .

3

$$T = \frac{2\pi}{5} = \frac{2\pi}{n} \Rightarrow n = 5$$

Let $x = A\cos(5t + \varepsilon)$, where $A, \varepsilon > 0$.

$$t = 0 \Rightarrow 1 = A\cos \varepsilon \quad -(1)$$

$$\dot{x} = -5A\sin(5t + \varepsilon)$$

$$\therefore -5\sqrt{3} = -5A\sin \varepsilon$$

$$t = 0 \Rightarrow A\sin \varepsilon = \sqrt{3} \quad -(2)$$

Solving (1) and (2) gives $A = 2, \varepsilon = \frac{\pi}{3}$

$$\therefore x = 2\cos\left(5t + \frac{\pi}{3}\right)$$

- (ii) Find the first time that the particle passes back through $x = 1$.

2

Let the first time be T , where $T > 0$

$$\therefore x(T) = x(0)$$

$$\therefore \cos\left(5T + \frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right)$$

$$\therefore 5T + \frac{\pi}{3} = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \dots$$

$$\therefore 5T + \frac{\pi}{3} = \frac{5\pi}{3} \quad (T > 0)$$

$$\therefore 5T = \frac{4\pi}{3}$$

$$\therefore T = \frac{4\pi}{15}$$

Question 12 (continued)

(d) Solve the inequality $\frac{x}{(x+1)(x-2)} \leq -\frac{1}{2}$

3

$$x \neq -1, 2$$

$$2(x+1)^2(x-2)^2 \times \frac{x}{(x+1)(x-2)} \leq -\frac{1}{2} \times 2(x+1)^2(x-2)^2$$

$$\therefore 2x(x+1)(x-2) \leq -(x+1)^2(x-2)^2$$

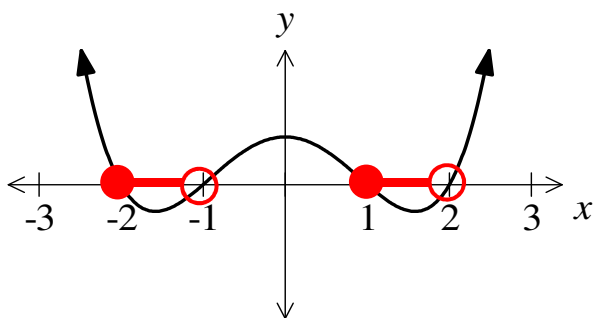
$$\therefore (x+1)^2(x-2)^2 + 2x(x+1)(x-2) \leq 0$$

$$\therefore (x+1)(x-2)[(x+1)(x-2) + 2x] \leq 0$$

$$\therefore (x+1)(x-2)(x^2 + x - 2) \leq 0$$

$$\therefore (x+1)(x-2)(x-1)(x+2) \leq 0$$

$$\therefore -2 \leq x < -1, 1 \leq x < 2$$

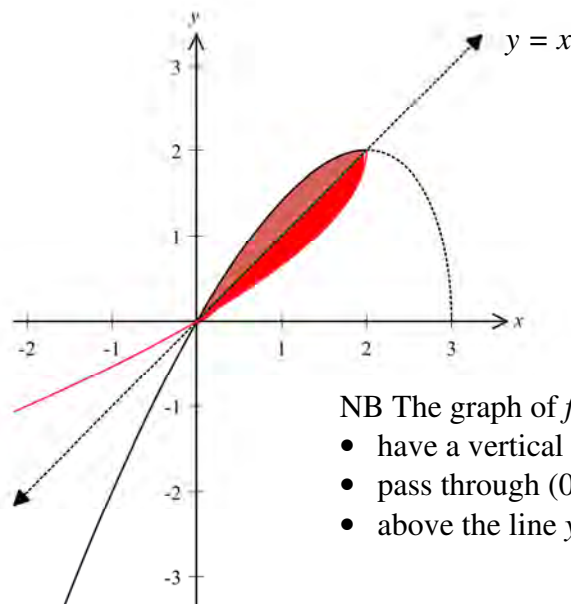


Question 13

- (a) The diagram below shows the graph of the curve $y = x\sqrt{3-x}$. The point P has coordinates $(2, 2)$ and is a stationary point. The function $f(x)$ is defined by $f(x) = x\sqrt{3-x}$, $x \leq 2$.

- (i) On the same diagram, sketch the region satisfied by $y \leq f(x)$ and $y \geq f^{-1}(x)$

3



- NB The graph of $f^{-1}(x)$, should
- have a vertical tangent at $(2, 2)$.
 - pass through $(0, 0)$.
 - above the line $y = x$ for $x < 0$

- (ii) Explain why the area A of the shaded region in (i) is given by

2

$$A = 2 \int_0^2 (x\sqrt{3-x} - x) dx$$

Do NOT attempt to evaluate this integral.

$y = f^{-1}(x)$ is a reflection of $y = f(x)$ in the line $y = x$.

So the area between $y = f(x)$ and $y = x$ for $0 \leq x \leq 2$ is the same as the area between $y = f^{-1}(x)$ and $y = x$ for $0 \leq x \leq 2$.

The area between $y = f(x)$ and $y = x$ for $0 \leq x \leq 2 = \int_0^2 (x\sqrt{3-x} - x) dx$

$$\therefore \text{Shaded area} = 2 \int_0^2 (x\sqrt{3-x} - x) dx$$

Question 13 (continued)

- (b) Nine different pies are to be divided between three people so that each person gets an odd number of pies. 2
Find the number of ways this can be done.

There are 3 cases

1. (1, 3, 5) with 3! different arrangements amongst the 3 people.
i.e. the first person gets 1 pie, the second person gets 3 pies
and the third gets 5 pies. This could be re-arranged in 3! ways.
 $\therefore {}^9C_1 \times {}^8C_3 \times 3!$
2. (1, 1, 7) with 3 different arrangements amongst the 3 people.
 $\therefore {}^9C_1 \times {}^8C_1 \times 3$
3. (3, 3, 3) with 1 possibilities
 $\therefore {}^9C_3 \times {}^6C_3$

$$\therefore {}^9C_1 \times {}^8C_3 \times 3! + {}^9C_1 \times {}^8C_1 \times 3 + {}^9C_3 \times {}^6C_3 = 4920$$

- (c) Show that $\lim_{x \rightarrow 5} \frac{x^3 - x^2 - 100}{x - 5} = 65$ 2

Let $f(x) = x^3 - x^2$.

$$f'(x) = 3x^2 - 2x$$

$$f(5) = 125 - 25 = 100$$

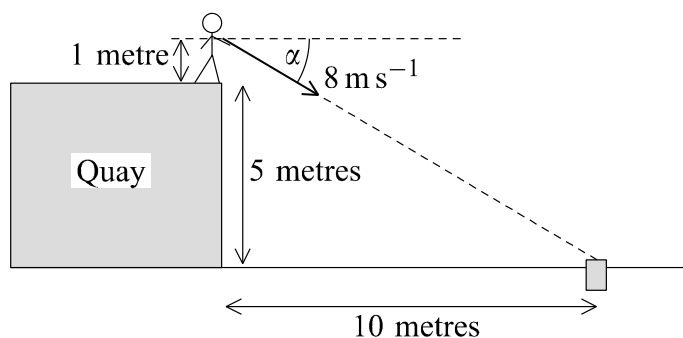
$$\begin{aligned} \therefore \lim_{x \rightarrow 5} \frac{x^3 - x^2 - 100}{x - 5} &= \lim_{x \rightarrow 5} \frac{f(x) - f(5)}{x - 5} \\ &= f'(5) \\ &= 3 \times 25 - 2 \times 5 \\ &= 65 \end{aligned}$$

Alternatively:

$$\begin{aligned} \therefore \lim_{x \rightarrow 5} \frac{x^3 - x^2 - 100}{x - 5} &= \lim_{x \rightarrow 5} \frac{x^3 - 125 - x^2 + 25}{x - 5} \\ &= \lim_{x \rightarrow 5} \frac{(x^3 - 125) - (x^2 - 25)}{x - 5} \\ &= \lim_{x \rightarrow 5} \left[(x^2 + 5x + 25) - (x + 5) \right] \\ &= 3 \times 25 - 10 \\ &= 65 \end{aligned}$$

Question 13 (continued)

(d)



$$x = 8t \cos \alpha \text{ and } y = 6 - 8t \sin \alpha - 4 \cdot 9t^2. \text{ (Do NOT prove this.)}$$

- (i) Find α . 2

Given that the initial velocity of the stone is directed at the can then

$$\tan \alpha = \frac{6}{10} = \frac{3}{5}$$

$$\therefore \alpha = 31^\circ$$

- (ii) Find the time that it takes for the stone to reach the level of the water. 2

The stone will reach the water when $y = 0$

$$\therefore 6 - 8t \sin \alpha - 4 \cdot 9t^2 = 0$$

$$\therefore 4 \cdot 9t^2 + 8t \sin \alpha - 6 = 0$$

$$\therefore 4 \cdot 9t^2 + 8t \times \frac{3}{\sqrt{34}} - 6 = 0 \quad \left[\tan \alpha = \frac{3}{5} \Rightarrow \sin \alpha = \frac{3}{\sqrt{34}} \right]$$

$$\therefore t = 0.76359, -1.60$$

$$\therefore t = 0.76$$

The stone will reach water level after 0.76 seconds.

- (iii) Find the distance between the stone and the can, when the stone hits the water. 2

The horizontal difference, d m, where $d = 10 - 8t \cos \alpha$

$$\therefore d = 10 - 8 \times 0.76 \times \frac{5}{\sqrt{34}} = 4.8$$

So the distance between the stone and the can is 4.8 m

Question 14

- (a) (i) Show that $\frac{1}{(k+2)!} - \frac{k+1}{(k+3)!} = \frac{2}{(k+3)!}$, where k is an integer. 1

$$\begin{aligned}\text{LHS} &= \frac{1}{(k+2)!} - \frac{k+1}{(k+3)!} \\ &= \frac{k+3-(k+1)}{(k+3)!} \\ &= \frac{2}{(k+3)!} = \text{RHS}\end{aligned}$$

- (ii) Prove by induction that, for all positive integers n , 3

$$\sum_{r=1}^n \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{n+1}}{(n+2)!}$$

Test $n = 1$:

$$\begin{aligned}\text{LHS} &= \sum_{r=1}^1 \frac{r \times 2^r}{(r+2)!} = \frac{1 \times 2^1}{(1+2)!} = \frac{2}{6} = \frac{1}{3} \\ \text{RHS} &= 1 - \frac{2^2}{(1+2)!} = 1 - \frac{4}{6} = \frac{1}{3}\end{aligned}$$

True for $n = 1$.

Assume true for $n = k$ i.e. $\sum_{r=1}^k \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{k+1}}{(k+2)!}$

Need to prove true for $n = k + 1$ i.e. $\sum_{r=1}^{k+1} \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{k+2}}{(k+3)!}$

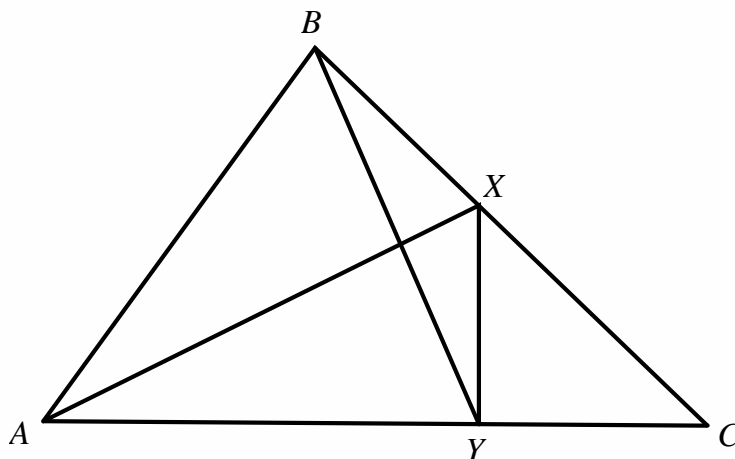
$$\begin{aligned}\text{LHS} &= \sum_{r=1}^{k+1} \frac{r \times 2^r}{(r+2)!} \\ &= \sum_{r=1}^k \frac{r \times 2^r}{(r+2)!} + \frac{(k+1) \times 2^{k+1}}{(k+3)!} \\ &= 1 - \frac{2^{k+1}}{(k+2)!} + \frac{(k+1) \times 2^{k+1}}{(k+3)!} \quad [\text{By assumption}] \\ &= 1 - 2^{k+1} \left[\frac{1}{(k+2)!} - \frac{k+1}{(k+3)!} \right] \\ &= 1 - 2^{k+1} \times \frac{2}{(k+3)!} \quad [\text{From (a) (i)}] \\ &= 1 - \frac{2^{k+2}}{(k+3)!} \\ &= \text{RHS}\end{aligned}$$

So the case $n = k + 1$ is true if the case $n = k$ is true.

So by the principle of mathematical induction, the formula is true for all positive integers.

Question 14 (continued)

- (b) X and Y are points on the sides BC and AC of a triangle ABC respectively such that $\angle AXC = \angle BYC$ and $BX = XY$.



- (i) Show that $\angle XAC = \angle YBC$. 2
 $\angle XAC + \angle AXC + \angle XCA = 180^\circ$ [angle sum $\triangle AXC$]
 $\therefore \angle XAC = 180^\circ - (\angle AXC + \angle XCA)$
 $= 180^\circ - \angle BYC - \angle BCY$ [data: $\angle AXC = \angle BYC$]
 $= \angle YBC$ [angle sum $\triangle YBC$]
- (ii) Hence, explain why $ABXY$ is a cyclic quadrilateral. 1
 As $\angle XAC = \angle YBC$ then $\angle XAY = \angle YBX$
 So $ABXY$ is a cyclic quadrilateral (converse of angles in the same segment)
- (iii) Prove that AX bisects the angle $\angle BAC$. 2
 $\angle BYX = \angle BAX$ (angles in same segment)
 Similarly, $\angle XBY = \angle XAY$.
 Now $\angle XBY = \angle BYX$ (equal angles opp. equal sides)
 $\therefore \angle XAY = \angle BAX$
 i.e. AX bisects $\angle BAC$

- (c) If $\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$, and $x = 70$ when $t = 0$, find x as a function of t . 3
- $$\frac{dt}{dx} = -\frac{1}{2}(x-6)^{-\frac{1}{2}}$$
- $$\therefore t = -(x-6)^{\frac{1}{2}} + C$$
- Substitute $x = 70, t = 0$
- $$\therefore 0 = -(70-6)^{\frac{1}{2}} + C$$
- $$\therefore C = 8$$
- $$\therefore t = -(x-6)^{\frac{1}{2}} + 8$$
- $$\therefore (x-6)^{\frac{1}{2}} = 8-t$$
- $$\therefore x = 6 + (8-t)^2$$

Question 14 (continued)

- (d) Liquid fuel is stored in a tank. At time t minutes, the depth of fuel in the tank is x cm. Initially there is a depth of 70 cm of fuel in the tank. There is a tap 6 cm above the bottom of the tank. The flow of fuel out of the tank is modeled by the differential equation

$$\frac{dx}{dt} = -2(x-6)^{\frac{1}{2}}$$

- (i) Explain what happens when $x = 6$. **1**
The fuel ceases to flow out the tap.
- (ii) Find how long it will take for the depth of fuel to fall from 70 cm to 22 cm. **2**
From (c), $x = 6 + (8-t)^2$
Find t , when $x = 22$.
 $22 = 6 + (8-t)^2$
 $\therefore (t-8)^2 = 16$
 $\therefore t = 8 \pm 4$
 $\therefore t = 4 \quad (0 \leq t \leq 8)$
It takes 4 seconds to fall to 22 cm.

End of solutions